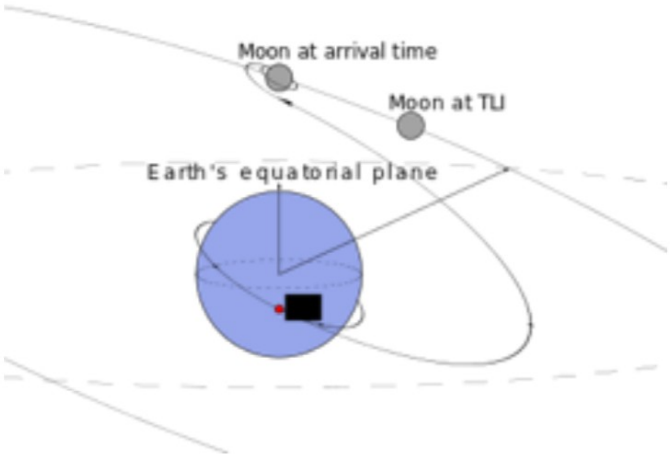
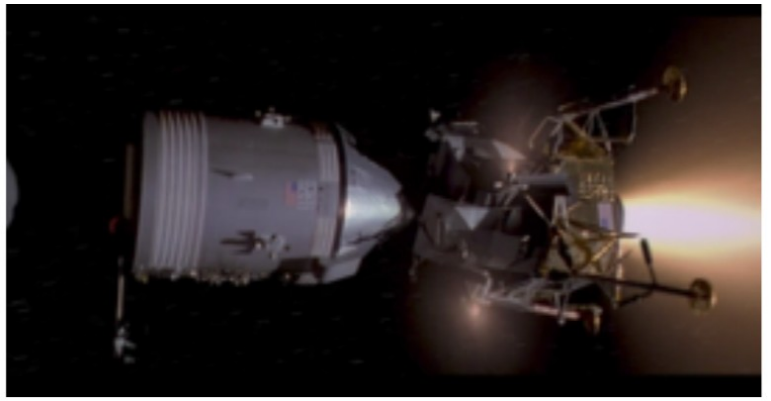
Lecture 10: Online Learning

Online learning example: to the moon (the first time)

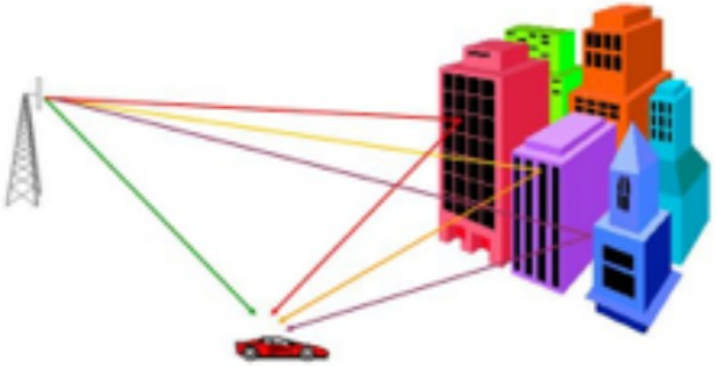
- Guidance computer predicts trajectories to the moon and back with
  - Noisy sensors
  - Imperfect models
  - Little computational power





Online learning -- recently

The original "Big Data"



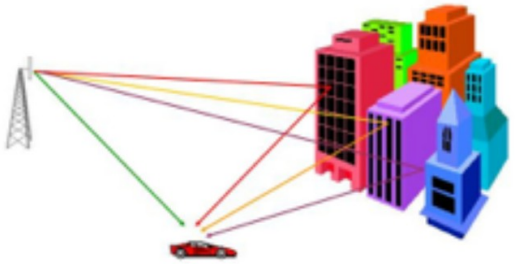
Wifi/cell-phones are *constantly* solving least squares to invert out multipath



Low power devices, high data rates

Online learning -- recently

The original "Big Data"



Wifi/cell-phones are *constantly* solving least squares to invert out multipath



Low power devices, high data rates



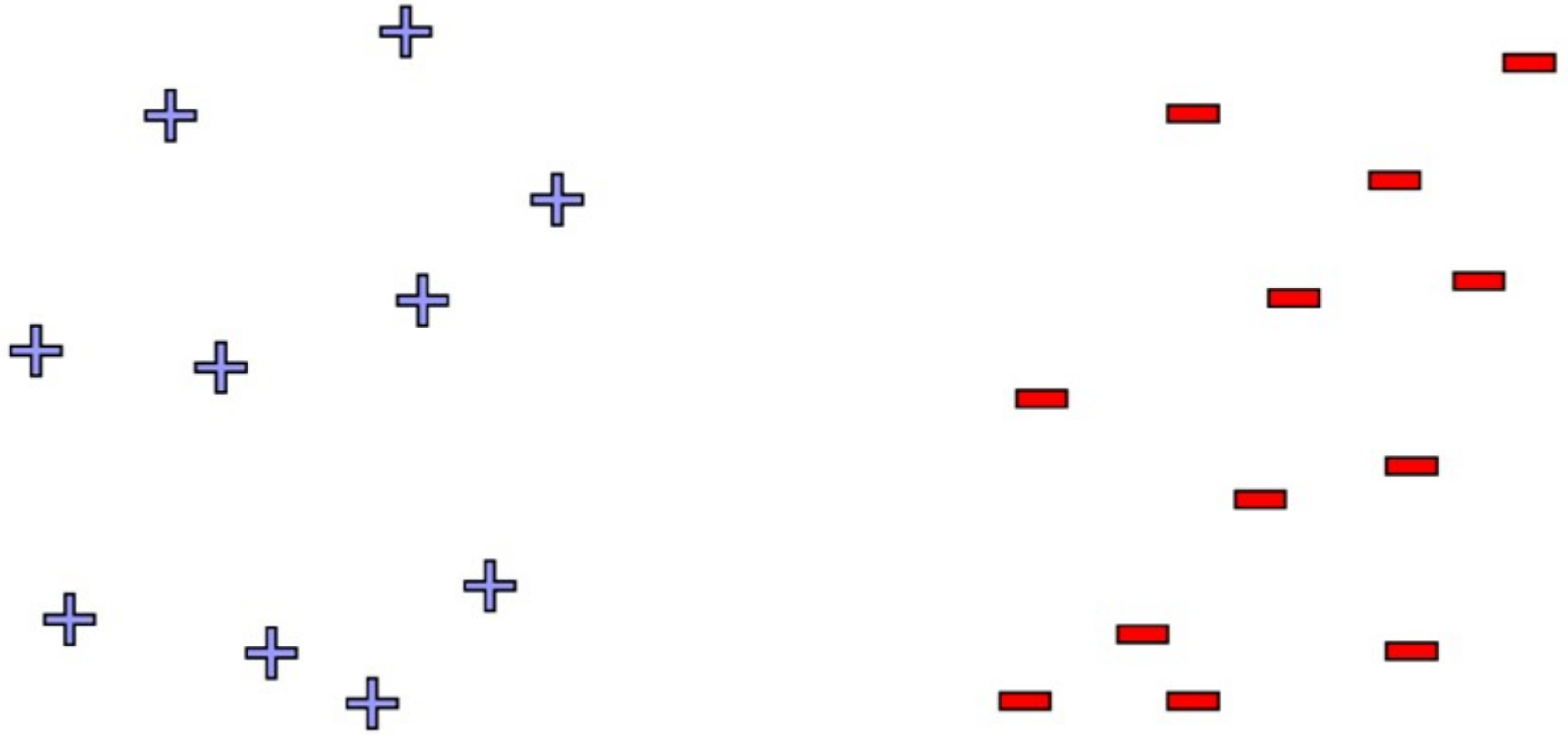
Gigabytes of data per second



Online learning

- Predictive ads deal with streaming data
  - User enters a query, and an ad must be selected
    - Observe the query, X_j, predict the ad – clicked or not clicked, Y_j!
  - User either clicks the ad or doesn't click the ad
    - Label on Y_j revealed afterwards
      - Google (or Facebook or whoever) gets a reward if the ad is clicked
  - Update the model for next time

Online classification

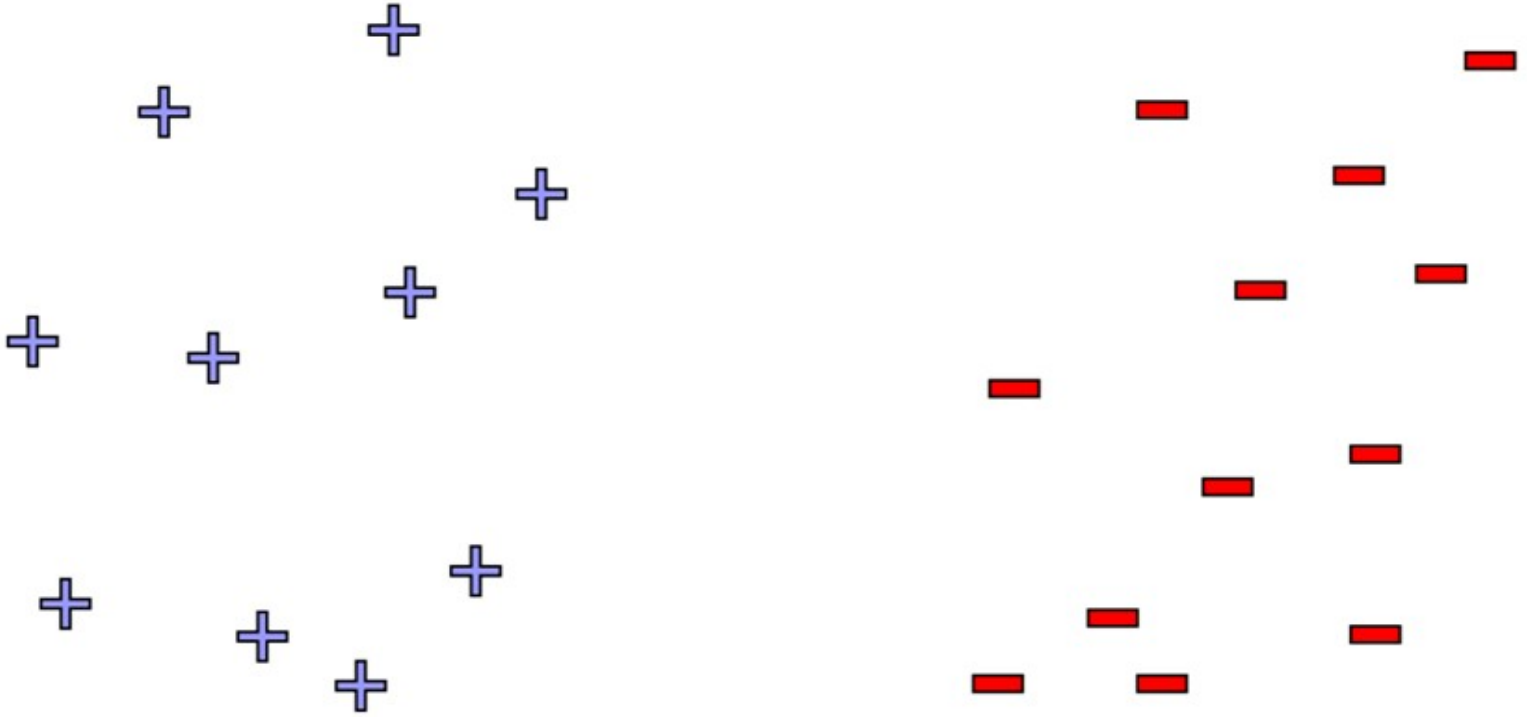


The perceptron algorithm

- Say y takes values {-1, 1}
- Linear model – prediction $y_k \sim \sin(x_k w + b)$
- Training:
  - Initialize weights (w_0, b_0)
  - At each step, k:
    - Observe features
    - Make predictions
    - Observe true class
    - Update model *if prediction is not equal to the truth*

Perceptron algorithm

- Perceptron is guaranteed to converge as long as the data is linearly separable



Perceptron algorithm

- The algorithm has that limitation, but is otherwise really cool!
  - No assumption about data distribution, *does not need to be i.i.d.*
  - Could be generated by an oblivious adversary
  - Makes a fixed number of mistakes, and it's done forever

    Even as you see infinite data
- But…perceptron is useless in practice
  - Real world data are not linearly separable
  - If it isn't separable, algorithm cycles forever
  - Even if separable, algorithm may not give good accuracy (may have small margin)